

CO4 - AT2 - Coding Golf Assesment

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QUESTIONS

Q1
Bubble Sort
25 marks

Q2
Insertion Sort
25 marks

Q3
Quick Sort
25 marks

Q4
Merger Sort
25 marks

4/4 answered

Q4 Merger Sort

25 marks

Write a C code for performing Merger Sort - n=8 A[] = {8 2 3 5 7 6 4 1}.

Your Code

```
1 #include <stdio.h>
2
3 void merge(int arr[], int l, int m, int r) {
4     int i, j, k;
5     int n1 = m - l + 1;
6     int n2 = r - m;
7     int L[n1], R[n2];
8     for (i = 0; i < n1; i++) L[i] = arr[l + i];
9     for (j = 0; j < n2; j++) R[j] = arr[m + 1 + j];
10    i = 0; j = 0; k = l;
11    while (i < n1 && j < n2) {
12        if (L[i] <= R[j]) {
13            arr[k] = L[i];
14            i++;
15        } else {
16            arr[k] = R[j];
17            j++;
18        }
19    }
20 }
```

Input

stdin input

Output



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Merger Sort
25 marks

4/4 answered

Q3 Quick Sorting

25 marks

Write a C code for performing Quick Sort - $n=7$ $A[] = \{8, 2, 3, 5, 7, 1, 6\}$.

Your Code

```
1 #include <stdio.h>
2
3 void swap(int* a, int* b) {
4     int t = *a;
5     *a = *b;
6     *b = t;
7 }
8
9 int partition(int arr[], int low, int high) {
10     int pivot = arr[high];
11     int i = low - 1;
12     for (int j = low; j < high; j++) {
13         if (arr[j] < pivot) {
14             i++;
15             swap(&arr[i], &arr[j]);
16         }
17     }
18     swap(&arr[i + 1], &arr[high]);
```

Input

stdin input

Output

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QUESTIONS

Q1
Bubble Sort
25 marks

Q2
Insertion Sort
25 marks

Q3
Quick Sort
25 marks

Q4
Merge Sort
25 marks

4/4 answered

Q2 Insertion Sorting

25 marks

Write a C code for performing Insertion Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5     int A[] = {8, 2, 3, 5, 7};
6     int i, j, key;
7
8     for (i = 1; i < n; i++) {
9         key = A[i];
10        j = i - 1;
11        while (j >= 0 && A[j] > key) {
12            A[j + 1] = A[j];
13            j--;
14        }
15        A[j + 1] = key;
16    }
17
18    for (i = 0; i < n; i++) {
```

Input

stdin input

Output

2 3 5 7 8

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QUESTIONS

Q1
Bubbled Sorting
25 marks

Q2
Insertion Sorting
25 marks

Q3
Quick Sorting
25 marks

Q4
Merger Sort
25 marks

4/4 answered

Q1 Bubbled Sorting

25 marks

Write a C code for performing Bubbled Sort - $n=5$ $A[] = \{8\ 2\ 3\ 5\ 7\}$.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int A[] = {8, 2, 3, 5, 7};
4     int n = 5;
5     for (int i = 0; i < n - 1; i++) {
6         for (int j = 0; j < n - i - 1; j++) {
7             if (A[j] > A[j + 1]) {
8                 int t = A[j];
9                 A[j] = A[j + 1];
10                A[j + 1] = t;
11            }
12        }
13    }
14    for (int i = 0; i < n; i++) {
15        printf("%d ", A[i]);
16    }
17    return 0;
18 }
```

Input

stdin input

Output

2 3 5 7 8